quantum theory and the many experiments that bear out its predictions, the quantum connection between two particles can persist even if they are on opposite sides of the universe. From the standpoint of their entanglement, notwithstanding the many trillions of miles of space between them, it's as if they are right on top of each other.

Numerous assaults on our conception of reality are emerging from modern physics; we will encounter many in the following chapters. But of those that have been experimentally verified, I find none more mind-boggling than the recent realization that our universe is not local.

The Red and the Blue

To get a feel for the kind of nonlocality emerging from quantum mechanics, imagine that Agent Scully, long overdue for a vacation, retreats to her family's estate in Provence. Before she's had time to unpack, the phone rings. It's Agent Mulder calling from America.

"Did you get the box—the one wrapped in red and blue paper?"

Scully, who has dumped all her mail in a pile by the door, looks over and sees the package. "Mulder, please, I didn't come all the way to Aix just to deal with another stack of files."

"No, no, the package is not from me. I got one too, and inside there are these little lightproof titanium boxes, numbered from 1 to 1,000, and a letter saying that you would be receiving an identical package."

"Yes, so?" Scully slowly responds, beginning to fear that the titanium boxes may somehow wind up cutting her vacation short.

"Well," Mulder continues, "the letter says that each titanium box contains an alien sphere that will flash red or blue the moment the little door on its side is opened."

"Mulder, am I supposed to be impressed?"

"Well, not yet, but listen. The letter says that *before* any given box is opened, the sphere has the capacity to flash either red or blue, and it